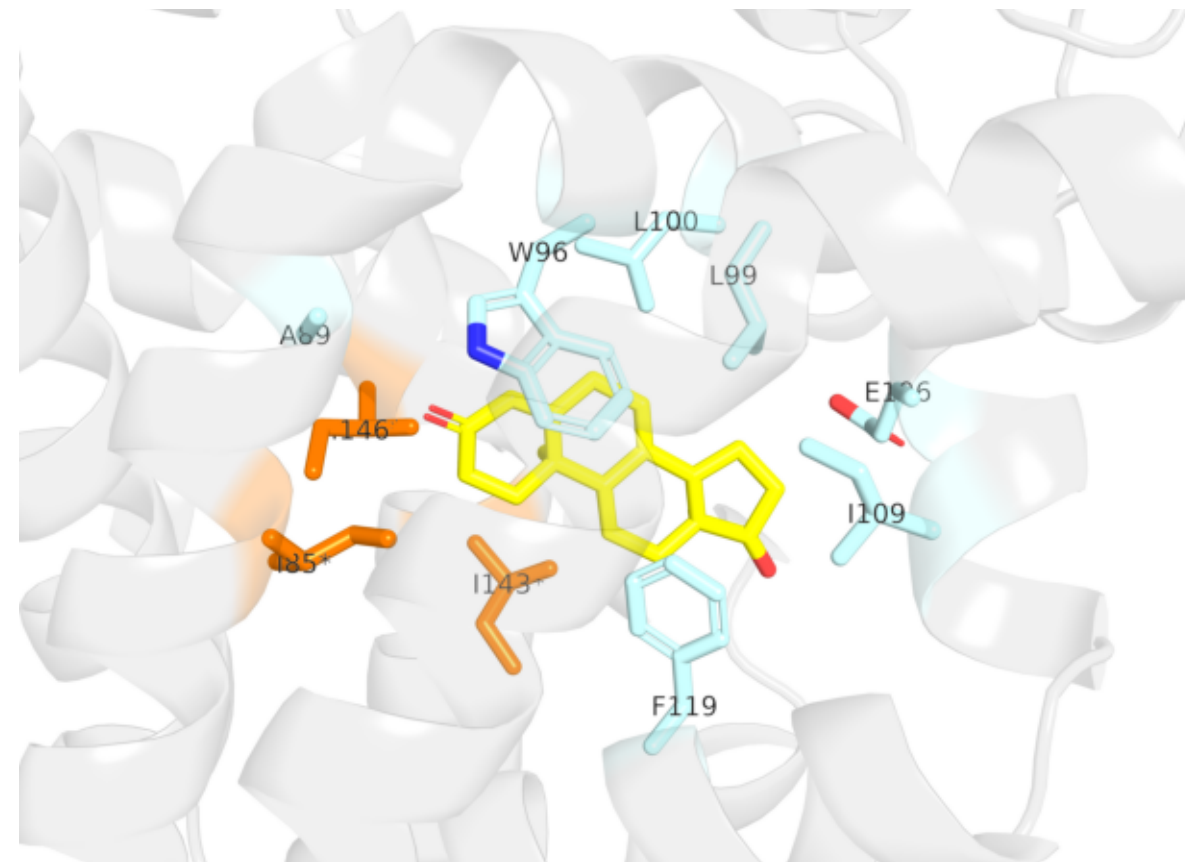


AcrR designed-pocket × ligand interactions (Boltz holo, 2-ligand homodimer; cognate protomer A)

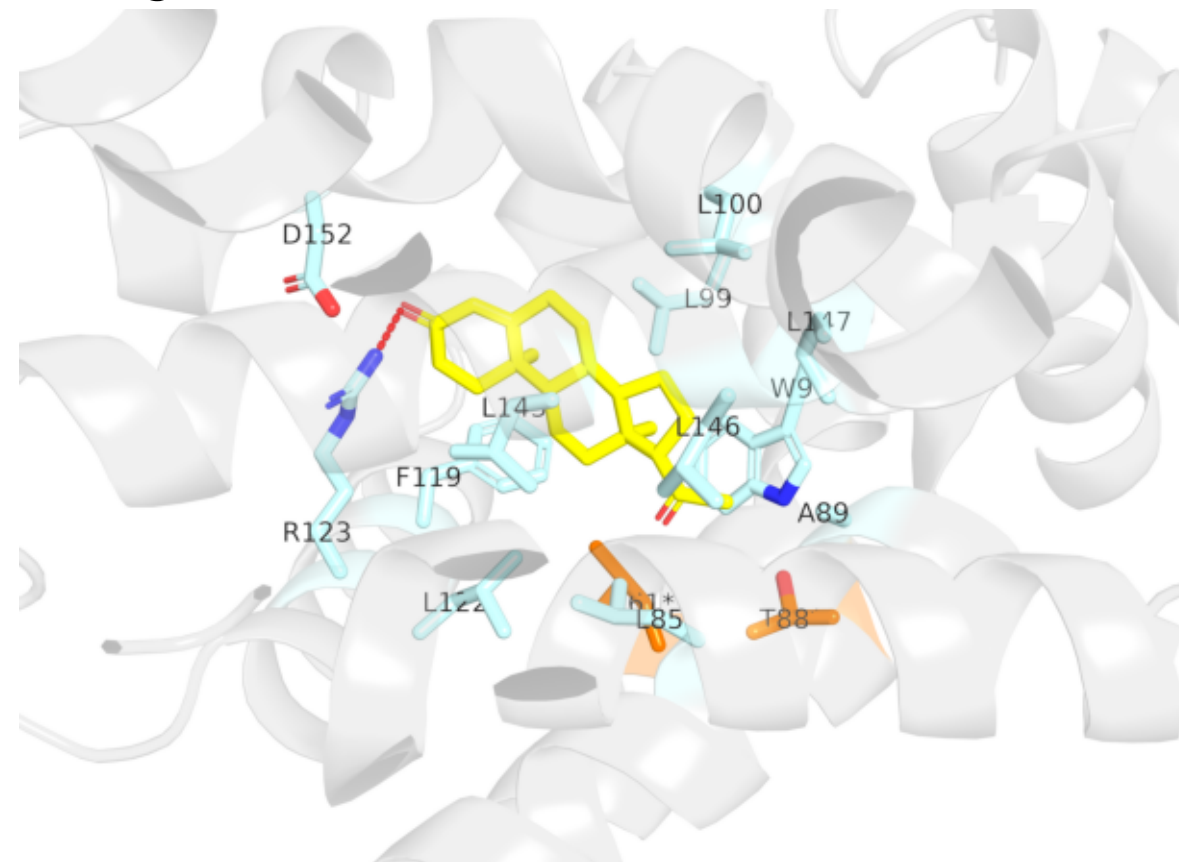
A Testosterone · des0039



mutations: I61L, L85I, L122F, L143I, L146I, L147F

⚠ **predicted A-ring FLIPPED — 3-keto 12 Å from R123 (D-ring bump: I85/I143/I146)**

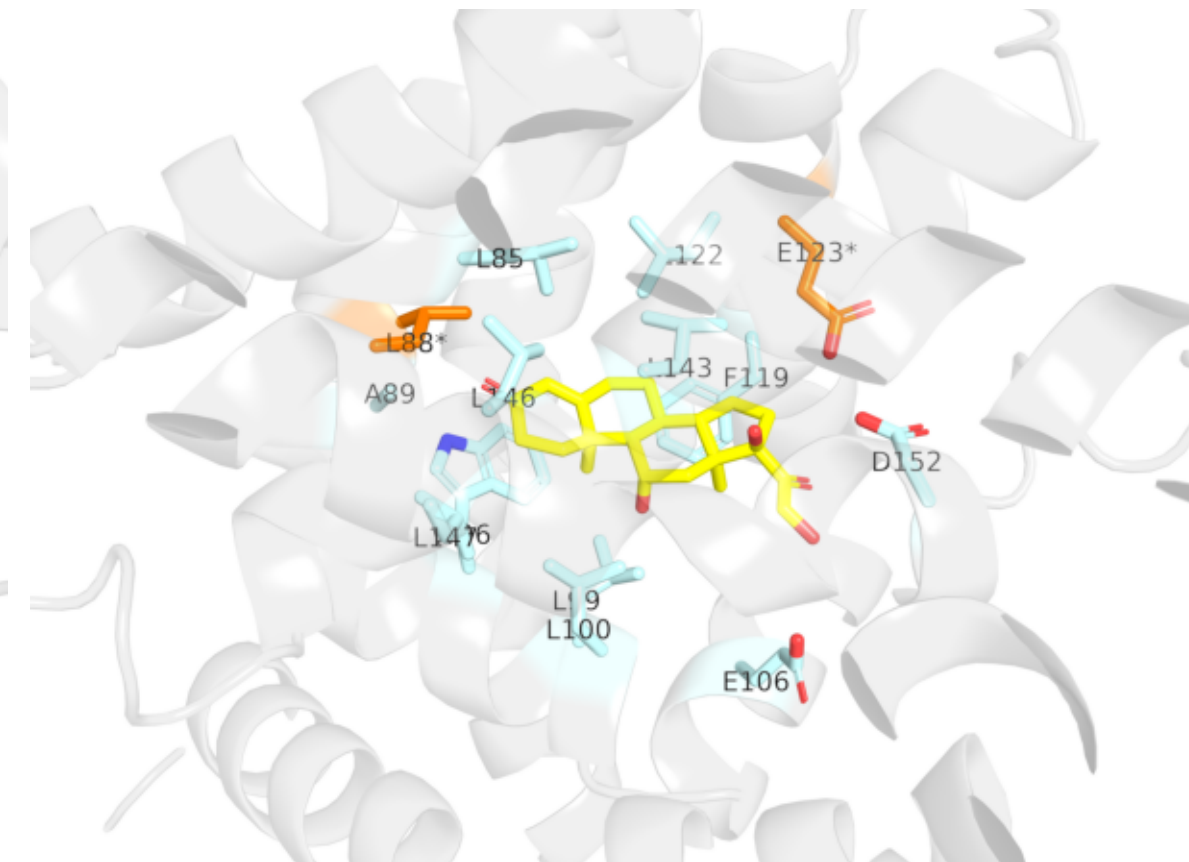
B Progesterone · des0002



mutations: I61L, Q88T

✓ **SAR-consistent — 3-keto 2.7 Å from R123; T88 reads C20 acetyl**

C Cortisol · des0001



mutations: I61V, Q88L, R123E

⚠ **3-keto buried — 12 Å from R123/E106; polyol poorly engaged**

Ligand = yellow sticks · binding-site residues (≤ 4.5 Å) = pale-cyan · design mutations = orange* · candidate H-bonds (≤ 3.5 Å) = red dashes · labels = MODEL numbering (exp = +9).
 ⚠ CAVEAT: de-novo Boltz poses — A-ring ORIENTATION is unreliable and flips (3-keto→R123: prog 2.7 Å ✓, test 12 Å ✗, cort 12 Å ✗). Recognition code (3-keto↔R123/E106) is SAR-supported; the POSES are not.